Probability 1 (MATH 11300) sample class – problem sheet

January 2021

Questions marked * are to be handed in.

*1. You arrive at a bus stop at 10 o'clock, knowing that the bus will arrive at some time uniformly distributed between 10 and 10:30.

   (a) What is the probability that you will have to wait longer than 10 minutes?

   (b) If at 10:15 the bus has not yet arrived, what is the probability that you will have to wait at least an additional 10 minutes?

2 A bus travels between the two cities A and B, which are 100 km apart. If the bus has a breakdown, the distance from the breakdown to city A has a uniform distribution over (0, 100). There is a bus service station in city A, in B, and at half way on the route between A and B. It is suggested that it would be more efficient to have the three stations located 25, 50, and 75 km, respectively, from A. Do you agree? Why? What would be the optimal location of the three stations?